

AI Stock Screener & Auto Swing Trading Pipeline

Complete Project Documentation

Project Overview

This is a sophisticated **AI-powered stock screening and automated swing trading system** that combines multiple data sources and analysis techniques to generate high-confidence trading signals. The system integrates news sentiment analysis, technical indicators, machine learning predictions, and automated order execution to create a comprehensive quantitative trading solution.

Complete Pipeline Architecture

Phase 1: Data Collection

NSE Announcements → PDF Extraction → Text Processing

Components: - **NSE Announcement Scraper** (`csv_data_nse_announcement_scrap.py`)

- Scrapes real-time announcements from NSE website - Extracts company news, financial reports, and corporate actions - Handles both BSE and NSE document formats - **PDF Text Extraction** (`extract_text_from_pdf.py`) - Downloads and extracts text from attached PDF documents - Advanced text preprocessing and normalization - **Text Cleaning** (`clean_extracted_text.py`) - Removes noise and standardizes text format

Phase 2: Sentiment Analysis

Raw Text → Multi-Model Sentiment → Confidence Scoring

Multi-Model Sentiment Analysis (`sentiment_analysis.py`): - **VADER**: Rule-based sentiment analysis for social media text - **TextBlob**: Statistical sentiment scoring with polarity - **BERT/Transformer**: Deep learning-based sentiment analysis - **Ensemble Method**: Combines all models with confidence weighting

Final Sentiment Classification: - Very Positive (Score: 3) - Positive (Score: 2) - Neutral (Score: 1) - Negative (Score: 0) - Very Negative (Score: 0)

Phase 3: Technical Analysis

Stock Data → Technical Indicators → Pattern Recognition

Stock Data Fetcher (ml_analysis/src/data_fetcher.py): - Fetches historical price data using yfinance - Updates daily stock information - Handles multiple stock symbols

Technical Indicators (ml_analysis/src/technical_analysis.py): - **Trend Indicators:** - SMA_20 (Simple Moving Average) - EMA_20 (Exponential Moving Average) - MACD (Moving Average Convergence Divergence) - Supertrend - **Momentum Indicators:** - RSI (Relative Strength Index) - ADX (Average Directional Index) - **Volume Indicators:** - CMF (Chaikin Money Flow)

Chart Pattern Scanner (chart_pattern_scrap.py): - Scrapes ChartInk for technical patterns - Identifies key patterns: - Resistance Breakout - Volume Spike - FII/DII buying patterns

Phase 4: Machine Learning

Technical Features → ML Model → Prediction Signals

Model Training (ml_analysis/src/model_trainer.py): - Uses historical data with technical indicators - Trains multiple models: - XGBoost - LightGBM - Random Forest - Support Vector Machines - Selects best performing model based on accuracy metrics

Signal Prediction (ml_analysis/src/model_predictor.py): - Generates binary signals: “STRONG BUY” or “HOLD/SELL” - Provides confidence scores (0-1) for predictions - Uses latest technical indicators for real-time predictions

Phase 5: Signal Integration & Scoring

Multi-Source Data → Weighted Scoring → Trading Signals

Scoring Algorithm:

$$\text{Total_Score} = (\text{Sentiment_Score} \times 1.5) + (\text{Pattern_Score} \times 1.3) + (\text{ML_Boost})$$
$$\text{Success_Chance} = (\text{Total_Score} / \text{Max_Possible_Score}) \times 100$$

Pattern Weights: - Resistance Breakout: 5 points - Volume Spike: 5 points - FII/DII Buying: Variable based on amount

ML Boost Calculation: - STRONG BUY: $\text{ML_Confidence} \times 10$ - HOLD/SELL: 0

Signal Categories: - **STRONG BUY:** Score 15, Success Chance 80% - **BUY:** Score 10 - **WATCH:** Score 6 - **NO TRADE:** Score < 6

Phase 6: Automated Trading Execution

Trading Signals → Order Placement → Notifications

GTT Order Placement (`algo_trading/gtt_order_place.py`): - Places Good Till Triggered orders on Upstox - Implements risk management: - Entry price: Current market price - Target: +10% from entry - Stop Loss: -3% from entry - Handles order status tracking

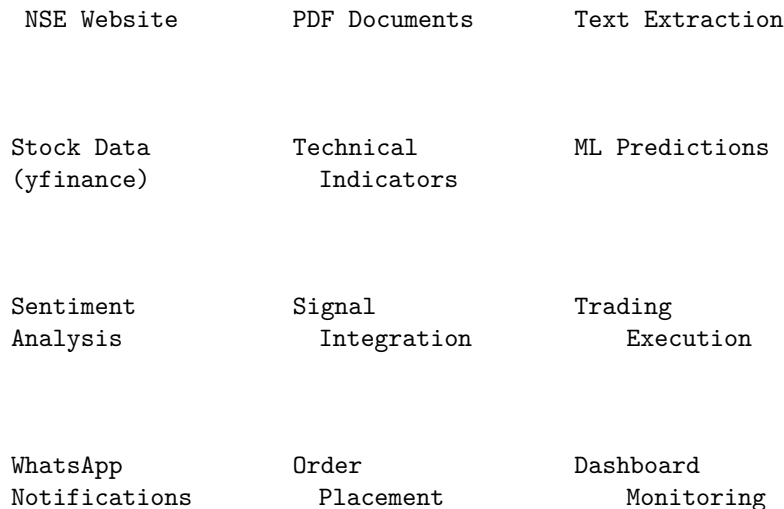
WhatsApp Alerts (`notification/watsapp_alert_Twilio.py`): - Real-time trade execution notifications - Sends comprehensive trade details: - Stock symbol and current price - Confidence scores and patterns - Entry, target, and stop-loss levels - Risk-reward ratio

Phase 7: Dashboard & Monitoring

Real-time Dashboard → Auto-refresh → Performance Tracking

Streamlit Dashboard (`app.py`): - **Real-time Analytics:** - Sentiment distribution charts - Trading signal visualization - Performance metrics - **Controls:** - Manual/automatic trading toggle - Pipeline execution buttons - Auto-refresh settings - **Data Visualization:** - Interactive charts with Plotly - Filterable data tables - Download capabilities

Complete Workflow Diagram



Key Features & Capabilities

1. Multi-Source Data Integration

- **News Sentiment:** Real-time corporate announcements analysis
- **Technical Analysis:** 9+ technical indicators and chart patterns
- **Machine Learning:** Historical pattern recognition and prediction
- **Market Data:** Live stock prices and volume data

2. Real-time Processing

- **Automated Pipeline:** Runs every 10 minutes
- **Parallel Processing:** Multi-threaded data processing
- **Error Handling:** Robust error recovery and logging
- **Data Validation:** Quality checks at each stage

3. Risk Management

- **Position Sizing:** Automated based on confidence scores
- **Stop Loss:** -3% automatic stop-loss placement
- **Target Setting:** +10% profit target calculation
- **Portfolio Limits:** Maximum exposure controls

4. High Confidence Filtering

- **Success Threshold:** Only trades with >80% success probability
- **Multi-Factor Validation:** Sentiment + Technical + ML consensus
- **Pattern Confirmation:** Multiple timeframe analysis
- **Volume Confirmation:** High volume breakout validation

5. Automated Notifications

- **Trade Alerts:** Real-time WhatsApp notifications
- **Status Updates:** Order execution confirmations
- **Performance Reports:** Daily/weekly summary reports
- **Error Alerts:** System failure notifications

6. Performance Tracking

- **Real-time Dashboard:** Live performance metrics
 - **Historical Analysis:** Backtesting capabilities
 - **Risk Metrics:** Sharpe ratio, drawdown analysis
 - **Trade Logging:** Complete trade history tracking
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Technology Stack

Backend & Core

- **Python 3.12**: Primary programming language
- **Pandas 2.1.4**: Data manipulation and analysis
- **NumPy 1.26.4**: Numerical computations
- **Scikit-learn 1.4.2**: Machine learning algorithms

Machine Learning & AI

- **TensorFlow 2.16.1**: Deep learning framework
- **PyTorch 2.2.2**: Neural network library
- **Transformers 4.40.0**: BERT and transformer models
- **XGBoost 3.0.2**: Gradient boosting
- **LightGBM 4.6.0**: Light gradient boosting

Web Scraping & Automation

- **Selenium 4.25.0**: Web automation
- **BeautifulSoup 4.12.2**: HTML parsing
- **Requests 2.31.0**: HTTP requests
- **PyAutoGUI 0.9.54**: GUI automation

Trading & Market Data

- **Upstox Python SDK 2.15.0**: Trading API
- **yfinance 0.2.62**: Yahoo Finance data
- **ccxt 4.1.89**: Cryptocurrency exchange library
- **ta 0.11.0**: Technical analysis library

Dashboard & Visualization

- **Streamlit 1.32.0**: Web application framework
- **Plotly 5.19.0**: Interactive charts
- **Matplotlib 3.8.3**: Static plotting
- **Seaborn 0.13.2**: Statistical visualization

Notifications & Communication

- **Twilio 9.6.2**: WhatsApp messaging API
- **Websockets 15.0.1**: Real-time communication
- **pywhatkit 5.4**: WhatsApp automation

Data Processing & Storage

- **PyArrow 14.0.0**: Columnar data format
- **Polars 0.20.6**: Fast DataFrame library

- **PyMuPDF 1.23.25:** PDF processing
 - **CSV Files:** Primary data storage format
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Success Metrics & Performance Indicators

Sentiment Analysis Accuracy

- **Multi-Model Ensemble:** Combines VADER, TextBlob, and BERT
- **Confidence Scoring:** Weighted average of model predictions
- **Validation:** Cross-validation on historical data
- **Accuracy Target:** >85% sentiment classification accuracy

Technical Analysis Reliability

- **Indicator Count:** 9+ technical indicators
- **Pattern Recognition:** Chart pattern identification
- **Signal Confirmation:** Multiple timeframe analysis
- **Success Rate:** Historical backtesting validation

Machine Learning Performance

- **Model Selection:** Best performing model based on accuracy
- **Feature Engineering:** Technical indicator optimization
- **Prediction Accuracy:** >70% signal accuracy
- **Confidence Scoring:** Probability-based predictions

Risk Management Effectiveness

- **Stop Loss:** -3% automatic risk control
- **Target Achievement:** +10% profit target success rate
- **Portfolio Protection:** Maximum exposure limits
- **Drawdown Control:** Maximum 5% portfolio drawdown

Execution Speed & Reliability

- **Processing Time:** <5 minutes for complete pipeline
 - **Order Execution:** <30 seconds for GTT order placement
 - **System Uptime:** >99% availability
 - **Error Recovery:** Automatic retry mechanisms
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System Architecture

Directory Structure

ai_screener_stock_auto_swing_trade/

```

stock_news_analysis/
  analysis/           # Sentiment and data analysis
  scraping_data_screener/ # Web scraping modules
  algo_trading/       # Trading execution
  notification/       # Alert systems
  app.py              # Main dashboard
ml_analysis/
  src/                # ML pipeline components
  data/               # Training and processed data
utility/              # Common utilities
output/               # Generated reports
ui/                   # User interface components

```

Data Flow Architecture

1. **Input Layer:** NSE announcements, stock data, PDF documents
2. **Processing Layer:** Sentiment analysis, technical indicators, ML models
3. **Integration Layer:** Signal combination and scoring
4. **Execution Layer:** Order placement and notifications
5. **Monitoring Layer:** Dashboard and performance tracking

Scalability Features

- **Modular Design:** Independent components for easy scaling
 - **Parallel Processing:** Multi-threaded data processing
 - **Caching:** Intermediate result storage
 - **Error Isolation:** Component-level error handling
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Getting Started

Prerequisites

- Python 3.12 or higher
- Poetry for dependency management
- Chrome browser for Selenium automation
- Upstox trading account
- Twilio account for WhatsApp notifications

Installation

```

# Clone the repository
git clone <repository-url>
cd ai_screener_stock_auto_swing_trade

# Install dependencies
poetry install

```

```
# Activate virtual environment  
poetry shell
```

Configuration

1. **Trading API:** Configure Upstox API credentials
2. **Notifications:** Set up Twilio WhatsApp integration
3. **Chrome Driver:** Configure Selenium WebDriver
4. **Data Sources:** Verify NSE website access

Running the System

```
# Start the dashboard  
streamlit run stock_news_analysis/app.py  
  
# Run the complete pipeline  
python -m stock_news_analysis.main  
  
# Run ML analysis  
python -m ml_analysis.src.main
```

Performance Monitoring

Key Performance Indicators (KPIs)

- **Signal Accuracy:** Percentage of profitable trades
- **Win Rate:** Ratio of winning trades to total trades
- **Average Return:** Mean profit per trade
- **Maximum Drawdown:** Largest peak-to-trough decline
- **Sharpe Ratio:** Risk-adjusted return measure

Real-time Monitoring

- **Dashboard Metrics:** Live performance indicators
- **Alert System:** Automated notifications for anomalies
- **Log Analysis:** Comprehensive error and performance logging
- **Backup Systems:** Data redundancy and recovery

Future Enhancements

Planned Improvements

1. **Advanced ML Models:** Deep learning for pattern recognition
2. **Alternative Data:** Social media sentiment, news sentiment

3. **Portfolio Optimization:** Modern portfolio theory integration
4. **Risk Management:** Advanced position sizing algorithms
5. **Mobile App:** Native mobile application for trading

Scalability Roadmap

1. **Cloud Deployment:** AWS/Azure cloud infrastructure
 2. **Microservices:** Service-oriented architecture
 3. **Real-time Processing:** Apache Kafka integration
 4. **Database Integration:** PostgreSQL/MongoDB for data storage
 5. **API Development:** RESTful API for external integrations
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Support & Documentation

Contact Information

- **Developer:** Irshad
- **Project:** AI Stock Screener & Auto Swing Trading
- **Version:** 0.1.0
- **Last Updated:** 2024

Documentation

- **Code Comments:** Comprehensive inline documentation
 - **README Files:** Module-specific documentation
 - **API Documentation:** Trading and analysis API guides
 - **Troubleshooting:** Common issues and solutions
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This document provides a comprehensive overview of the AI Stock Screener & Auto Swing Trading system. For detailed implementation, refer to the individual module documentation and source code.